

Vitoria Aquino Guardieiro

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Education

Ph.D. in Computer and Information Sciences

University of Pennsylvania

Aug 2024 – Now

Philadelphia, USA

- Advisor: Eric Wong

M.Sc. in Mathematical Modeling

Getulio Vargas Foundation (FGV)

Mar 2021 – Jun 2023

Rio de Janeiro, Brazil

- Advisors: Jorge Poco (FGV) and Marcos M. Raimundo (State University of Campinas)
- Dissertation: Multi-objective Optimization in Machine Learning - A Systematic Review

B.Sc. in Applied Mathematics

Getulio Vargas Foundation (FGV)

Feb 2017 – Dec 2020

Rio de Janeiro, Brazil

- Advisors: Jorge Poco (FGV) and Marcos M. Raimundo (State University of Campinas)
- Bachelor's dissertation: Application of multi-objective optimization in machine learning problems to decrease discrimination

Publications

Manuscripts under review

V. Guardieiro, A. Khare, A. Stein, E. Wong. "Instruction following by boosting attention of large language models". Under review at NeurIPS 2026. Earlier versions presented at NeurIPS 2025 Workshop MechInterp (spotlight), ICLR 2026 Workshop Sci4DL, and ICLR 2026 Workshop Trustworthy AI.
<https://arxiv.org/abs/2506.13734>

V. Guardieiro, F. L. Usberti, J. Poco, F. J. V. Zuben, M. M. Raimundo. "An efficient box-based scalarization algorithm for non-convex multi-objective optimization". Under review at European Journal of Operational Research.

Peer-reviewed publications

G. Valdrighi, A. M. Ribeiro, Jansen SB Pereira, V. Guardieiro, A. Hendricks, D. M. Filho, J. D. N. Garcia, F. F. Bocca, T. B. Veronese, L. Wanner, M. M. Raimundo. (2025). "Best practices for responsible machine learning in credit scoring". *Neural Computing and Applications*. <https://doi.org/10.1007/s00521-025-11520-y>

P. Silva, V. Guardieiro, B. Barr, C. Silva, and L. G. Nonato. (2025). "Visagreement: Visualizing and Exploring Explanations (Dis)Agreement". *IEEE Transactions on Visualization and Computer Graphics*.
<https://doi.org/10.1109/TVCG.2025.3558074>

V. Guardieiro, F.I. de Oliveira, H. Doraiswamy, L.G. Nonato, and C. Silva. (2024). "TopoMap++: A faster and more space efficient technique to compute projections with topological guarantees". *IEEE Transactions on Visualization and Computer Graphics*. <https://doi.org/10.1109/TVCG.2024.3456365>

P. Solunke, V. Guardieiro, J. Rulff, P. Xenopoulos, G. Y. Y. Chan, B. Barr, L. G. Nonato, and C. Silva. (2024). "Mountaineer: Topology-Driven Visual Analytics for Comparing Local Explanations". *IEEE Transactions on Visualization and Computer Graphics*. <https://doi.org/10.1109/TVCG.2024.3418653>

V. Guardieiro, M. M. Raimundo, and J. Poco. (2023). "Enforcing fairness using ensemble of diverse Pareto-optimal models". *Data Mining and Knowledge Discovery*.
<https://doi.org/10.1007/s10618-023-00922-y>

V. Guardieiro, M. M. Raimundo, and J. Poco. (2022). “Analyzing the Equity of the Brazilian National High School Exam by Validating the Item Response Theory’s Invariance”. *Proceedings of the 15th International Conference on Educational Data Mining*. <https://doi.org/10.5281/zenodo.6852946>

Research Experience

Research Scientist Oct 2023 – Jul 2024
New York University *New York, US*

- Research on Topological Data Analysis for Machine Learning explainability at the Visualization Imaging and Data Analysis Center (VIDA) with Professor Claudio Silva

Visiting Researcher Jan 2023 – Apr 2023
New York University *New York, US*

- Visited the Visualization Imaging and Data Analysis Center (VIDA) and worked with Professor Claudio Silva on using Topological Data Analysis for comparing machine learning explanation methods

Undergraduate Research Assistant 2017 – 2020
Getulio Vargas Foundation *Rio de Janeiro, Brazil*

- Worked with Waze data, developing a clustering algorithm for interval time-spatial structures. Advised by Prof Jorge Poco
- Studied Natural Language Processing, contributing to databases such as Open Multilingual Wordnet (OMW) and Suggested Upper Merged Ontology (SUMO). Advised by Prof Alexandre Rademaker
- Elaboration of the Consumer Price Index - Web (IPC-Web) for the Brazilian Institute of Economics (IBRE). Advised by Prof Renato Rocha Souza

Other Experience

Teaching Assistant Fall 2025 – Spring 2026
University of Pennsylvania *Philadelphia, USA*

- Disciplines CIS 7000 – Accelerating Research with Generative AI (Fall 2025) and CIS 1990 – Introduction to Designing LLM Agents (Spring 2026)

Teaching Assistant 2018, 2021 – 2022
Getulio Vargas Foundation *Rio de Janeiro, Brazil*

- Disciplines from the B.Sc. in Applied Mathematics and the B.Sc. in Data Science: Machine Learning (2022/1 and 2022/2), Continuous Optimization (2022/2), Optimization for Data Science (2022/2), Statistical Inference (2021/2), and Introduction to Computing (2018/1)

Data Science Intern/Data Scientist May 2020 – Jul 2021
Gabriel Tecnologia *Rio de Janeiro, Brazil*

- Gabriel is a SoftBank-funded AI startup that develops technology focused on public security
- I developed computer vision object detection, classification, and tracking models for the automatic detection of anomalies in security camera videos (Python, Pytorch)

IT Intern Jan 2020 – Apr 2020
BTG Pactual *Rio de Janeiro, Brazil*

- BTG Pactual is the biggest investment bank in Latin America
- I developed dashboards, visualizations, and metrics for internal IT products (SQL, C++)

Mathematical Modelling Intern Jan 2019 – Sep 2019
PSR - Energy Consulting and Analytics *Rio de Janeiro, Brazil*

- PSR is a global provider of technological solutions in the areas of electricity and natural gas (E&G)
- I developed optimization programs for maintenance scheduling in thermoelectric and hydroelectric plants (Julia, JuMP)

Awards & Scholarships

Coordination for the Improvement of Higher Education Personnel (CAPES)

M.Sc. Fellowship

Mar 2021 – Feb 2023

FGV's Center for the Development of Mathematics and Science (CDMC)

B.Sc. Fellowship

2017 – 2020

Brazilian Mathematical Olympiad of Public Schools (OBMEP)

One gold medal, two silver medals, and three bronze medals (Top 0.01%)

2011 – 2016

Jovens Gênio Algar (JoGAR) - Coding Marathon

First place in the advanced category (2016), third place (2015), and fourth place (2014)

2014 – 2016

BeeCrowd (previously URI Online Judge) - Competitive Programming Platform

Top 0.2% of users

Specialized Skills

Programming Languages: Python, Julia, MATLAB, R, C, C++, Javascript

Languages: English (TOEFL 112/120), Portuguese (Native)